Deliverable 1: Single-Segment Optimization

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1 Problem and notation

Minimize J = integral from 0 to 1 of $[P'(t)]^2$ dt, subject to P(0)=0 and P(1)=1. Time and displacement are fixed. Squaring the speed penalizes high-speed peaks. This cost is not a direct measure of electrical energy.

Write P(t)=p1*t+p2*t^2+...+pn*t^n. P(0)=0 removes the constant term. With coefficients ordered as p=[p1,...,pn]^T, the quadratic program is:

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minimize p^T Q p, subject to A p = b
Q_ij = i*j/(i+j-1), i,j = 1,...,n
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The expression for Q follows by expanding $P'(t)^2$ and integrating each power of t.

2 Linear polynomial: Question 1

P(t)=p1*t and P(1)=1 give p1=1. Thus P(t)=t and J=1.

3 Quadratic polynomial: Question 2

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p = [p1,p2]^T
Q = [[1,1],[1,4/3]], A = [1,1], b = 1
J = p1² + 2*p1*p2 + (4/3)*p2²
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The constraint gives p1=1-p2. Substitution gives J=1+$p2^2$/3. The minimum is at p2=0: p=[1,0]^T, P(t)=t, J=1.

4 Cubic polynomial: Question 3

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p = [p1,p2,p3]^T
Q = [[1,1,1],[1,4/3,3/2],[1,3/2,9/5]]
A = [1,1,1], b = 1
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For L=p^T Qp+lambda^T(Ap-b), stationarity and feasibility give the Karush-Kuhn-Tucker equations:

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[2Q  A^T] [p     ] = [0]
[A   0  ] [lambda]   [b]
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The solution is p=[1,0,0]^T and lambda=-2. Direct substitution gives Ap=1 and 2Qp+A^T*lambda=0. Again P(t)=t and J=1.

5 Extra terminal velocity: Question 4

Now also require P'(1)=-2. The trajectory must reach position 1 while moving in the negative direction. P(t)=t is no longer feasible.

Table 1. Exact solutions with the extra constraint

Quantity	Quadratic	Cubic
A	[[1,1],[1,2]]	[[1,1,1],[1,2,3]]
b	[1,-2]^T	[1,-2]^T
Optimal p	[4,-3]^T	[1/4,9/2,-15/4]^T
P(t)	$4t-3t^2$	$t/4+9t^2/2-15t^3/4$
J	4	17/8 = 2.125

For the quadratic, p1+p2=1 and p1+2p2=-2 give p1=4 and p2=-3. Integrating $(4-6t)^2$ gives J=4.

For the cubic, set p3=c. The constraints give p2=-3-2c and p1=4+c. Substitution into the cost gives:

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J(c) = 4+c+(2/15)*c²
J'(c) = 1+(4/15)*c = 0  =>  c = -15/4
J''(c) = 4/15 > 0
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Therefore p=[1/4,9/2,-15/4]^T. Checking the endpoints gives P(1)=1/4+9/2-15/4=1 and P'(1)=1/4+9-45/4=-2. Exact integration gives J=17/8.

6 Why the original solution is a straight line

Cauchy-Schwarz gives integral$[P'(t)^2]$dt >= (integral P'(t)dt)^2 = 1. Equality requires constant velocity. On this unit interval, P(t)=t. The Euler-Lagrange equation gives the same result, P''(t)=0.

Increasing degree cannot reduce the original cost below 1. With the extra terminal velocity, however, the cubic has useful freedom and reduces the cost from 4 to 17/8. Higher degree helps when it improves a constrained solution, not automatically.

Q is positive definite: every nonzero coefficient vector gives a nonzero derivative with positive squared integral. Hence each feasible finite-dimensional problem has a unique minimum. The archived verify_theory.py script checks the constraints and integrals with exact rational arithmetic.

Source: MIT VNAV Lab4, individual Deliverable 1. https://vnav.mit.edu/labs_2023/lab4/exercises.html